

# Coding & Solution

## 1. Serverless Gateway Handler (`index.py`)

The serverless execution starts at root-level `index.py`. It adds the development folder path to `sys.path`, loads the environmental variables, imports the global Flask instance, and maps Vercel serverless request inputs dynamically:

## 2. Database Initialization Logic

The database manager utilizes safe SQLite column migrations, allowing columns to be altered dynamically without breaking older schemas:

- Alters users table to add name, role, last\_login, and status.
- Seeds default demo credentials with secure scrypt hashing.
- Ensures read-only environments run cleanly by catching write errors.